

Lab Overview: Microsoft Access Forms and Reports

This lab will help you learn how to design, create, and customize forms and reports in Microsoft Access through practical exercises.

Part 1: Database Setup

Exercise 1: Create a Sample Database

1. Open Microsoft Access and create a new blank database named "FormsReportsLab.accdb"
2. Create the following tables:

Customers Table:

- CustomerID (AutoNumber, Primary Key)
- FirstName (Text)
- LastName (Text)
- Email (Text)
- Phone (Text)
- JoinDate (Date/Time)

Products Table:

- ProductID (AutoNumber, Primary Key)
- ProductName (Text)
- Category (Text)
- UnitPrice (Currency)
- InStock (Yes/No)

Orders Table:

- OrderID (AutoNumber, Primary Key)
- CustomerID (Number, Foreign Key)
- OrderDate (Date/Time)
- TotalAmount (Currency)
- Status (Text)

OrderDetails Table:

- OrderDetailID (AutoNumber, Primary Key)
- OrderID (Number, Foreign Key to Orders.OrderID)
- ProductID (Number, Foreign Key to Products.ProductID)
- Quantity (Number)
- UnitPrice (Currency) - This captures the price at the time of order

3. Create relationships:

- Customers to Orders: One-to-Many
 - Connecting CustomerID in both tables
- Orders to OrderDetails: One-to-Many
 - Connecting OrderID in both tables
- Products to OrderDetails: One-to-Many
 - Connecting ProductID in both tables

4. Add at least 5 sample records to each table

Customers Table

CustomerID	FirstName	LastName	Email	Phone	JoinDate
1	John	Smith	john.smith@email.com	555-123-4567	01/15/2024
2	Sarah	Johnson	sarah.j@email.com	555-234-5678	02/20/2024
3	Michael	Williams	m.williams@email.com	555-345-6789	03/10/2024
4	Emily	Brown	emily.brown@email.com	555-456-7890	04/05/2024
5	David	Jones	david.jones@email.com	555-567-8901	05/12/2024
6	Jennifer	Garcia	j.garcia@email.com	555-678-9012	06/25/2024
7	Robert	Miller	robert.m@email.com	555-789-0123	07/08/2024
8	Jessica	Davis	jessica.d@email.com	555-890-1234	08/17/2024
9	Christopher	Rodriguez	c.rod@email.com	555-901-2345	09/03/2024
10	Amanda	Martinez	a.martinez@email.com	555-012-3456	10/14/2024

Products Table

ProductID	ProductName	Category	UnitPrice	InStock
1	Laptop 15" Pro	Electronics	\$1299.99	Yes
2	Wireless Mouse	Computer Accessories	\$29.99	Yes
3	External Hard Drive 1TB	Computer Accessories	\$89.99	Yes
4	Bluetooth Speaker	Electronics	\$59.99	Yes
5	Wireless Headphones	Electronics	\$149.99	Yes
6	USB-C Cable Pack	Computer Accessories	\$19.99	Yes
7	Monitor 27" 4K	Electronics	\$349.99	Yes
8	Smartphone Case	Phone Accessories	\$24.99	Yes
9	Tablet 10"	Electronics	\$499.99	Yes
10	Smart Watch	Electronics	\$199.99	No
11	Wireless Charger	Phone Accessories	\$39.99	Yes
12	Keyboard Mechanical	Computer Accessories	\$79.99	No
13	Office Software	Software	\$149.99	Yes
14	Antivirus Software	Software	\$59.99	Yes
15	Graphics Card	Computer Components	\$399.99	No

Orders Table

OrderID	CustomerID	OrderDate	TotalAmount	Status
1	3	03/15/2024	\$1329.98	Completed
2	1	04/02/2024	\$209.97	Completed
3	5	05/20/2024	\$1799.97	Completed
4	2	06/10/2024	\$149.99	Completed
5	8	07/05/2024	\$599.97	Completed
6	4	07/22/2024	\$89.99	Completed
7	7	08/15/2024	\$449.98	Completed
8	10	09/03/2024	\$279.98	Shipped
9	6	09/18/2024	\$1349.98	Processing
10	9	10/10/2024	\$1939.95	Pending
11	2	10/15/2024	\$199.99	Pending
12	5	10/20/2024	\$79.98	Pending
13	3	10/25/2024	\$549.98	Processing

OrderDetails Table

OrderDetailID	OrderID	ProductID	Quantity	UnitPrice	Subtotal
1	1	1	1	\$1299.99	\$1299.99
2	1	2	1	\$29.99	\$29.99
3	2	6	1	\$19.99	\$19.99
4	2	8	2	\$24.99	\$49.98
5	2	11	1	\$39.99	\$39.99
6	3	7	1	\$349.99	\$349.99
7	3	9	1	\$499.99	\$499.99
8	3	15	1	\$399.99	\$399.99
9	3	13	1	\$149.99	\$149.99

10	3	14	1	\$59.99	\$59.99
11	4	5	1	\$149.99	\$149.99
12	5	9	1	\$499.99	\$499.99
13	5	8	4	\$24.99	\$99.98
14	6	3	1	\$89.99	\$89.99
15	7	7	1	\$349.99	\$349.99
16	7	2	1	\$29.99	\$29.99
17	7	6	3	\$19.99	\$59.97
18	8	4	2	\$59.99	\$119.98
19	8	11	4	\$39.99	\$159.96
20	9	1	1	\$1299.99	\$1299.99
21	9	5	1	\$149.99	\$149.99
22	10	1	1	\$1299.99	\$1299.99
23	10	7	1	\$349.99	\$349.99
24	10	5	1	\$149.99	\$149.99
25	10	3	1	\$89.99	\$89.99
26	10	8	2	\$24.99	\$49.98
27	11	10	1	\$199.99	\$199.99
28	12	6	4	\$19.99	\$79.96
29	13	9	1	\$499.99	\$499.99
30	13	11	2	\$39.99	\$79.98

Part 2: Basic Forms

Exercise 2: Create a Simple Form

1. Select the Customers table in the Navigation Pane
2. Click the Create tab, then click Form
3. Save the form as "CustomerForm"
4. Test navigation between records
5. Add a new record using the form

When you create this form using the Form button on the Create tab, Access will automatically generate:

- All the fields from your Customers table
- A form header area
- Standard navigation buttons at the bottom (First Record, Previous Record, Next Record, Last Record, New Record)

Exercise 3: Form Wizard

1. Click the Create tab, then click Form Wizard
2. Create a form for the Products table including all fields
3. Choose a Columnar layout
4. Save as "ProductsWizardForm"
5. Test the form functionality

Part 3: Advanced Forms

Exercise 4: Create a Form with Subform

1. Use Form Wizard to create a form for Customers with a subform for Orders
2. Include CustomerID, FirstName, and LastName from Customers
3. Include OrderID, OrderDate, and TotalAmount from Orders
4. Choose a Datasheet layout for the subform
5. Save as "CustomerOrdersForm"
6. Test the relationship functionality

Exercise 5: Customizing Forms in Design View

1. Open "CustomerForm" in Design View
2. Add a calculated control showing the number of days since the customer joined
 - In the Design tab, click Text Box
 - Set the Control Source property to: =DateDiff("d",[JoinDate],Date())
 - Set the Caption to "Days as Customer"
3. Save and test the form

Exercise 6: Creating a Form with Validation

1. Create a new form based on the Products table
2. In Design View, add validation:
 - For UnitPrice field, set Validation Rule to >0 and Validation Text to "Price must be greater than zero"
 - Add a Required property for ProductName

3. Add a combo box for Category with predefined values
4. Save as "ProductValidationForm"
5. Test by entering valid and invalid data

Part 4: Custom Buttons and Controls

Exercise 6: Adding Command Buttons

1. Open "CustomerForm" in Design View
2. Add the following buttons using the Button control:
 - New Record button
 - Delete Record button
 - Print Form button
3. Use the Command Button Wizard to set actions
4. Save and test functionality

Part 5: Basic Reports

Exercise 7: Create a Simple Report

1. Select the Customers table
2. Click Create tab, then Report
3. Save as "CustomerReport"
4. View the report in Print Preview
5. Export to PDF

Exercise 8: Report Wizard

1. Click Create tab, then Report Wizard
2. Create a report showing Products grouped by Category
3. Include ProductName, UnitPrice, and InStock fields
4. Choose a stepped layout and portrait orientation
5. Save as "ProductsByCategory"
6. View in different views